ChArUco ceramic calibration target

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Overview

The ChArUco ceramic calibration target uses lithographic deposition of blue chrome on an alumina (Al2O3) substrate, achieving feature tolerances of +/-2 um. Designed for metrology-grade applications, semiconductor inspection, and high-accuracy stereo calibration.

Available sizes

Part number	Board size	Grid	Square	Marker	Dictionary	Tolerance
CHA-C-6X6-100x100-10	100 x 100 mm	9 x 9	10 mm	7 mm	DICT_6X6_250	+/-2 um
CHA-C-6X6-150x150-15	150 x 150 mm	9 x 9	15 mm	11 mm	DICT_6X6_250	+/-2 um
CHA-C-6X6-200x200-20	200 x 200 mm	9 x 9	20 mm	15 mm	DICT_6X6_250	+/-2 um

Substrate

- Material: 96% alumina ceramic (Al2O3)
- Thickness: 3 mm
- Surface: lithographed blue chrome on white ceramic
- CTE: approximately 7 um/(m.K)
- Operating temperature: -40 to +150 C
- Flatness: <= 0.05 mm across full board

Feature accuracy

- Feature tolerance: +/-2 um (lithographic process)
- Edge definition: < 5 um
- Suitable for: stereo calibration, structured light, interferometric setups

Deliverables

Each target ships with:

- Dimensional inspection report (included)
- Datasheet PDF
- OpenCV JSON configuration
- Kalibr YAML configuration

ILAC-recognised calibration certificate available as an add-on (SVC-X-CERT-ILAC).

Storage and handling

Handle with lint-free gloves. Store in the supplied hard case. Clean with isopropanol and a lint-free cloth. Do not use abrasive materials.